

ARAD, Romania

WMO: 152000

Lat: 46.1336N

Lon: 21.3536E

Elev: 387

StdP: 14.49

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.1	15.6	6.5	7.8	12.3	11.5	10.0	17.4	21.5	33.4	17.8	33.1	3.2	0	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.5	93.8	69.9	90.8	69.6	87.6	68.7	73.4	87.1	71.6	85.4	70.1	83.2	5.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.5	110.1	78.4	67.7	103.4	76.3	66.1	97.4	74.4	37.2	86.2	35.6	84.4	34.3	82.3	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.0	14.0	11.7	DB	2.3	97.6	7.9	3.4	-3.4	100.1	-8.0	102.1	-12.5	104.0	-18.2	106.5	(4)
(5)			WB	1.9	77.0	7.8	3.1	-3.7	79.2	-8.2	81.0	-12.6	82.7	-18.3	84.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.0	32.0	35.3	43.3	53.7	62.2	68.7	72.1	72.3	63.4	53.7	44.1	34.2		
	DBStd	16.22	8.56	9.76	8.27	7.06	6.53	6.34	5.80	6.01	6.86	8.09	8.38	8.77		
	HDD50	2013	559	413	233	44	2	0	0	0	57	213	492			
	HDD65	5102	1024	831	672	342	131	35	8	10	110	358	626	955		
	CDD50	3111	0	2	27	154	381	561	686	690	402	170	37	1		
	CDD65	727	0	0	0	2	46	146	230	235	62	6	0	0		
	CDH74	8240	0	0	0	67	579	1619	2601	2626	674	73	1	0		
	CDH80	3328	0	0	0	7	124	610	1170	1206	206	5	0	0		
	WSAvg	4.6	4.6	5.1	5.5	5.2	4.6	4.1	4.0	3.8	4.1	4.3	4.9	4.6		
	PrecAvg	24.8	1.4	1.4	1.5	2.0	2.4	3.2	2.8	2.2	2.2	2.0	1.9	1.9		
(15) Precipitation	PrecMax	33.2	2.8	3.5	5.4	5.2	5.7	7.1	6.3	6.2	5.5	4.3	4.4	4.1		
	PrecMin	11.5	0.3	0.0	0.2	0.1	0.7	0.8	0.9	0.1	0.4	0.0	0.0	0.1		
	PrecStd	5.2	0.7	0.9	1.1	1.1	1.1	1.5	1.5	1.4	1.1	1.2	1.0	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.5	61.5	70.0	80.4	86.4	93.5	98.3	98.5	91.4	79.8	69.7	57.2		
		MCWB	48.8	51.0	54.5	62.7	66.4	71.2	70.4	70.3	67.2	62.5	57.8	50.6		
	2%	DB	49.9	56.1	65.3	75.3	83.5	89.9	93.7	94.4	85.3	75.8	64.1	52.6		
		MCWB	45.6	48.3	52.1	59.3	65.4	70.7	70.4	69.4	64.9	60.9	54.8	47.9		
	5%	DB	46.6	51.8	61.1	71.5	80.5	86.8	90.4	90.9	81.4	71.7	60.4	48.9		
		MCWB	43.6	46.1	50.4	56.9	64.2	69.2	69.8	69.4	64.2	59.6	53.1	45.9		
	10%	DB	43.2	48.0	57.0	67.7	77.0	83.3	86.6	87.1	77.3	67.1	56.7	45.6		
		MCWB	41.2	44.1	48.3	55.1	62.7	67.6	68.7	68.3	62.8	57.7	51.4	43.4		
	0.4%	WB	49.6	52.0	56.8	63.9	69.4	75.9	75.5	75.1	69.4	64.8	58.7	51.7		
		MCDB	53.6	59.1	67.2	76.6	81.3	87.6	90.0	89.1	84.5	75.8	66.5	56.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	46.3	49.3	53.6	60.9	67.2	72.8	73.3	72.7	67.1	62.5	56.3	48.7		
		MCDB	49.1	55.3	62.3	72.7	79.9	85.4	87.1	87.4	80.5	71.9	62.5	52.0		
	5%	WB	43.7	46.9	51.2	58.8	65.4	70.8	71.3	71.0	65.5	60.6	54.0	46.1		
		MCDB	46.2	51.1	59.4	69.2	77.3	82.8	85.1	85.2	77.4	69.1	59.4	48.8		
	10%	WB	41.2	44.4	49.1	56.5	63.6	68.9	69.8	69.3	63.8	58.6	51.6	43.5		
		MCDB	43.0	47.7	56.2	65.2	74.6	80.3	83.2	82.6	74.2	66.3	56.1	45.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	14.4	18.7	21.2	22.1	22.1	23.5	24.0	21.3	20.1	15.5	10.7		
		MCDBR	15.1	20.5	25.3	27.8	27.6	27.5	28.8	29.1	27.5	25.4	20.5	14.6		
		MCWBR	11.2	13.8	14.7	14.4	12.8	11.6	10.3	10.2	11.0	13.0	12.8	11.0		
		MCDBR	14.2	18.1	22.3	24.6	25.1	25.1	25.7	25.4	23.7	22.5	19.1	13.7		
	5% WB	MCWBR	11.0	12.6	13.7	13.4	12.6	11.5	10.7	10.8	10.9	12.4	12.6	10.8		
(40) Clear-Sky Solar Irradiance	taub		0.339	0.365	0.402	0.458	0.440	0.437	0.450	0.456	0.430	0.390	0.361	0.339		
	taud		2.330	2.267	2.201	2.058	2.174	2.220	2.185	2.171	2.209	2.310	2.346	2.347		
	Ebn at Noon		235	251	258	253	263	264	259	252	246	239	224	220		
	Edh at Noon		25	32	39	49	45	44	45	44	39	30	24	22		
(44) All-Sky Solar Radiation	RadAvg		359	629	1030	1471	1784	1955	1937	1739	1213	816	474	282		
	RadStd		40	93	137	140	164	152	150	162	133	86	41	41		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.17	N/A	N/A	N/A	N/A	N/A	-216	N/A	+178	N/A
(47)	Regional (63 neighbors)	+1.24	N/A	N/A	N/A	+0.68	N/A	-220	-331	+209	+106

Nomenclature: See separate page